
Finite- K Posterior Learning with One-Draw Primal Updates

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Abstract

1 Multi-sample reasoning generates several candidate solutions and selects among
2 them at inference time. Reasoning models use training objectives defined for a
3 single solver execution, although deployment evaluates a candidate pool under a
4 fixed budget. We place a regularized posterior over solver parameters, using each
5 posterior sample as one candidate solver. For a prescribed pool size K and rea-
6 soning depth, we minimize a differentiable surrogate of the probability that all K
7 candidates fail. An unbiased stochastic estimate of this finite- K surrogate requires
8 K solver executions per update. To address this, we derive an exact Fenchel rep-
9 resentation of the surrogate whose primal update uses one posterior draw given
10 a per-problem dual state. We also track this state with a closed-form Bregman
11 update and one running failure estimate per problem. Together, this optimizes
12 the finite- K surrogate using one posterior draw per update. Across challenging
13 reasoning benchmarks, we validate finite- K posterior learning improves multi-
14 sample performance.

15 1 Introduction

16 2 Related work

17 2.1 Multi-sample Inference and Test-time Scaling

18 Multi-sample inference improves performance by generating multiple candidate solutions and ag-
19 gregating their complementary outcomes. Its benefit was quantified in code generation through
20 Pass@ K , which measures whether at least one sampled program passes the tests [1]. For reasoning
21 tasks, self-consistency aggregates diverse reasoning paths through answer-level voting [2], while out-
22 come and process verifiers rank candidate solutions using learned correctness signals [3, 4]. Struc-
23 tured search methods further extend this principle by expanding and pruning intermediate reasoning
24 branches instead of sampling complete solutions independently [5]. These inference procedures
25 can be viewed as forms of test-time scaling that allocate additional compute to broader candidate
26 exploration. Repeated-sampling studies demonstrate that increasing the number of attempts can sub-
27 stantially expand problem coverage [6], whereas systematic analyses show that the resulting gains
28 depend on the inference algorithm, selector, model size, and allocation of compute across prob-
29 lems [7, 8]. Collectively, this literature establishes the sampling budget K as an important inference-
30 time resource, but generally leaves the candidate-generating distribution unchanged during training.

31 This train-inference mismatch has motivated inference-aware learning methods that optimize the
32 collective utility of multiple samples. Recent approaches directly optimize K -sample objectives
33 using reinforcement learning [9, 10], tailor fine-tuning to best-of- N selection [11], or modify train-
34 ing to preserve coverage under repeated generation [12]. These methods align a point policy with

multi-sample inference, but their updates either operate on groups of output trajectories or exploit autoregressive likelihood structure. Our work instead learns a posterior $q_\phi(\theta)$ over solver parameters while retaining the selector of the underlying reasoning method. We optimize a differentiable surrogate of the finite- K all-fail risk and introduce a Fenchel reformulation with lagged per-problem dual states, such that one posterior draw yields an unbiased gradient for the fixed-dual primal subproblem. This provides a parameter-space formulation of inference-aware learning without executing all K solvers in every primal update.

2.2 Posterior learning for reasoning

Researchers have used posterior distributions over model parameters to represent uncertainty and instantiate multiple neural networks from a single learned distribution. Bissiri et al. generalize Bayesian updating by combining task losses with KL regularization toward a prior, while loss-calibrated inference adapts posterior approximations to downstream decision utilities [13–15]. Stochastic variational methods made neural weight distributions trainable through sampled parameters and backpropagation [16, 17], and natural-gradient methods later scaled this approach to modern deep networks [18, 19]. Post-hoc alternatives such as Laplace approximations and SWAG also construct sampleable weight distributions around trained solutions [20, 21]. These methods primarily optimize the expected loss of an individual sampled model or target predictive uncertainty and calibration; their posterior objectives do not account for the collective success of a fixed-size candidate pool.

Recent reasoning methods have begun to use stochastic computation to explore multiple solution paths. GRAM learns distributions over latent reasoning trajectories, whereas PTRM injects noise into recursive latent states without learning a parameter posterior [22, 23]. More closely related approaches move exploration into parameter space: PSN-RLVR perturbs policy weights with an adaptive noise schedule, and 3PO samples one or more policies from an IVON posterior to diversify RLVR rollouts and advantage groups [24, 25]. Although 3PO also learns a distribution over reasoning-model parameters, its generalized variational objective remains linear in the posterior through an expected GRPO loss, and it uses posterior samples to improve training-time exploration. Our method directly trains a solver-parameter posterior for the deployed inference protocol by minimizing the probability that all K posterior-sampled solvers fail, while retaining the prescribed reasoning depth and selector.

2.3 Efficient optimization of multi-sample objectives

Training objectives that evaluate several sampled candidates together are expensive because each update must normally generate the entire candidate set. Direct estimators evaluate at least K samples for a group of size K , even when they reuse subsets of a larger batch to reduce variance [10, 26]. Another line of work reduces this cost by carrying information from earlier updates in a running estimate of the average sample outcome. SCGD introduced this approach for objectives that first average random outcomes and then score the average [27]. NASA later used a single-loop design to track both this average and the model gradient [28]. These methods replace a fresh multi-sample estimate at every update with a state that is refined over time.

For finite training sets, later methods maintain a separate running estimate for each example. SOX updates the estimates associated with the current minibatch and uses their previous values to form the model update, allowing each update to use only a few new samples [29]. ALEXR develops a Fenchel-based method for convex problems, updating a helper variable from one sample batch and the model from an independent batch [30]. For smooth scoring functions, its Bregman update reduces to an ordinary moving average. In our setting, the running state summarizes the recent failure rate of posterior-sampled solvers under fixed- K inference, while the selector remains unchanged. We fix one scalar per problem before the current solver draw, use that draw for the posterior update, and update the scalar afterward. This order gives an unbiased gradient for the update with the scalar held fixed, while our tracking analysis quantifies how sampling noise and posterior changes affect the scalar and the finite- K surrogate gap.

85 3 Finite- K posterior learning

86 We formulate the exact all-fail risk under the prescribed inference protocol and obtain a differ-
 87 entiable finite- K surrogate for posterior learning (section 3.1). Fenchel conjugacy introduces a
 88 per-problem dual state that makes the surrogate’s posterior-dependent term a one-draw expectation
 89 (section 3.2). We update this state from running failure estimates through a Bregman update, and
 90 its tracking gap measures the error from using a lagged dual state in the surrogate objective (sec-
 91 tion 3.3).

92 3.1 Posterior learning for fixed sampling budgets

93 Many reasoning problems admit multiple solution paths, so a single solver execution can fail even
 94 when another succeeds. Multi-sample inference uses a fixed budget of K executions to increase
 95 the chance that the candidate pool contains a successful solution. We make the candidate-generating
 96 distribution trainable by learning a posterior $q_\phi(\theta)$ over solver parameters, regularized toward a prior
 97 $p_0(\theta)$. At inference, we draw K parameter samples from this posterior and run each sampled solver
 98 at the prescribed reasoning depth D . The selector associated with the underlying reasoning method
 99 then chooses or aggregates their outputs. We retain that selector and its established training protocol
 100 to isolate posterior learning from selector design. Learning a new selector or co-training the selector
 101 and posterior is a complementary research direction. Our objective targets candidate coverage. The
 102 retained selector determines which candidate the system returns.

103 For problem x , let $F(\theta, x) \in \{0, 1\}$ denote the exact failure indicator for solver θ at depth D , with
 104 $F = 1$ for failure. For independent draws $\theta_{1:K} \stackrel{\text{iid}}{\sim} q_\phi$,

$$\begin{aligned} \Pr(\text{all } K \text{ solvers fail} \mid x) &= \mathbb{E}_{\theta_{1:K}} \left[\prod_{k=1}^K F(\theta_k, x) \right] \\ &= (\mathbb{E}_{\theta \sim q_\phi} [F(\theta, x)])^K. \end{aligned} \quad (1)$$

105 The product is the all-fail indicator. Independence yields the K -th power, so minimizing it maxi-
 106 mizes Pass@ K .

107 This all-fail probability induces the regularized exact target

$$\min_{\phi} \mathbb{E}_x \left[(\mathbb{E}_{\theta \sim q_\phi} [F(\theta, x)])^K \right] + \tau_{\text{KL}} D_{\text{KL}}(q_\phi \| p_0). \quad (2)$$

108 The data term is the expected all-fail probability, and the KL term regularizes the solver posterior
 109 toward p_0 [13, 14].

110 The discrete indicator F does not provide a pathwise gradient through the solver. We therefore
 111 replace it during training with a differentiable failure score $f(\theta, x) \in [0, 1]$ and write $\bar{f}_\phi(x) :=$
 112 $\mathbb{E}_{\theta \sim q_\phi} [f(\theta, x)]$ for its posterior mean. The resulting differentiable training objective is

$$\min_{\phi} \mathbb{E}_x [\bar{f}_\phi(x)^K] + \tau_{\text{KL}} D_{\text{KL}}(q_\phi \| p_0). \quad (3)$$

113 For $K = 1$, its data term reduces to the expected surrogate failure of one posterior draw. When
 114 $f = F$, the optimization problems in eqs. (2) and (3) coincide; otherwise, eq. (3) serves as a smooth
 115 proxy for the exact all-fail risk.

116 3.2 One-draw min-max posterior learning

117 The product of K independent failure scores is an unbiased estimator of $\bar{f}_\phi(x)^K$, but computing it
 118 requires K solver executions per update. A one-draw alternative raises one sampled failure score to
 119 the K -th power. This quantity estimates $\mathbb{E}_{\theta \sim q_\phi} [f(\theta, x)^K]$, whereas the finite- K surrogate contains
 120 $(\mathbb{E}_{\theta \sim q_\phi} [f(\theta, x)])^K$. We therefore seek an exact reformulation of the surrogate in eq. (3) that avoids
 121 K solver executions at every update.

122 The finite- K surrogate has an exact min-max form. This form makes the posterior-dependent update
 123 a one-draw expectation for a fixed dual state.

124 Define $\psi_K(\lambda) = (K-1)(\lambda/K)^{K/(K-1)}$ on $[0, K]$. For $K \geq 2$ and $r \in [0, 1]$,

$$r^K = \max_{\lambda \in [0, K]} \{\lambda r - \psi_K(\lambda)\}, \quad \lambda^*(r) = Kr^{K-1}. \quad (4)$$

125 Applying eq. (4) to $r = \bar{f}_\phi(x)$ gives the equivalent saddle objective

$$\min_{\phi} \max_{\{\lambda_x \in [0, K]\}_x} \left\{ \mathbb{E}_x [\mathbb{E}_{\theta \sim q_\phi} [\lambda_x f(\theta, x)] - \psi_K(\lambda_x)] + \tau_{\text{KL}} D_{\text{KL}}(q_\phi \| p_0) \right\}. \quad (5)$$

126 For a finite training set, eq. (5) requires one scalar dual state λ_x for each training problem.

127 For a fixed current value of λ_x , its posterior-dependent term is an ordinary expectation over $\theta \sim q_\phi$.
 128 Assume that q_ϕ admits a differentiable reparameterization and that differentiation can pass through
 129 the expectation. One posterior draw then gives an unbiased stochastic gradient of the fixed-dual
 130 term. The optimal dual state is $\lambda_x^* = K \bar{f}_\phi(x)^{K-1}$. If $\lambda_x = \lambda_x^*$, the one-draw posterior gradient
 131 equals the finite- K surrogate data gradient. Otherwise, it remains unbiased for the current fixed-
 132 dual objective, whereas it need not equal the gradient of the nonlinear surrogate objective. The KL
 133 regularizer remains part of the posterior update.

134 3.3 Bregman update for the dual state

135 The optimal dual state depends on $\bar{f}_\phi(x)$ and changes as the posterior changes. Re-estimating it
 136 from multiple posterior draws at every update would remove the one-draw advantage. Instead, we
 137 update the dual state from the current sampled failure score $f_t = f(\theta_t, x)$.

138 Let $D_{\psi_K}(\lambda, \mu) = \psi_K(\lambda) - \psi_K(\mu) - \psi'_K(\mu)(\lambda - \mu)$ denote the Bregman divergence generated by
 139 ψ_K . We apply the Bregman-proximal ascent step

$$\lambda_{x,t+1} = \arg \max_{\lambda \in [0, K]} \left\{ \lambda f_t - \psi_K(\lambda) - \eta^{-1} D_{\psi_K}(\lambda, \lambda_{x,t}) \right\}, \quad \eta > 0. \quad (6)$$

140 Since $\psi'_K(\lambda) = (\lambda/K)^{1/(K-1)}$, this update has a closed form. With $\beta = 1/(1 + \eta)$ and $\hat{f}_{x,t-1} =$
 141 $\psi'_K(\lambda_{x,t})$,

$$\begin{aligned} \hat{f}_{x,t} &= \beta \hat{f}_{x,t-1} + (1 - \beta) f_t, \\ \lambda_{x,t+1} &= K \hat{f}_{x,t}^{K-1}. \end{aligned} \quad (7)$$

142 The closed form reduces dual ascent to a scalar exponential moving average (EMA). It requires nei-
 143 ther an inner optimization loop nor an additional solver execution. A larger β suppresses sampling
 144 noise but increases lag when the posterior failure rate changes.

145 At update t , we compute $\lambda_{x,t} = K \hat{f}_{x,t-1}^{K-1}$ from the running estimate available before drawing θ_t . We
 146 hold this state fixed during the posterior update. Afterward, we incorporate f_t into $\hat{f}_{x,t}$ and use the
 147 result at the next update. This order fixes the dual state before the current posterior sample, which
 148 gives the conditional one-draw gradient property in section 3.2.

Algorithm 1 One-draw finite- K posterior learning

Require: Posterior q_ϕ , prior p_0 , pool size K , coefficient β , and failure estimates $\{\hat{f}_x\}$

- 1: **for** each training update t **do**
 - 2: Set and detach $\lambda_{x,t} \leftarrow K \hat{f}_{x,t-1}^{K-1}$ for each minibatch problem x
 - 3: Draw $\theta_t \sim q_\phi$ once and compute $f(\theta_t, x)$
 - 4: Update ϕ with $\lambda_{x,t} f(\theta_t, x)$ and the KL regularizer while holding $\lambda_{x,t}$ fixed
 - 5: Update $\hat{f}_{x,t} \leftarrow \beta \hat{f}_{x,t-1} + (1 - \beta) f(\theta_t, x)$
 - 6: **end for**
-

149 Direct finite- K training uses K solver executions per update. The exact min-max reformulation
 150 makes the posterior update a one-draw expectation once the current dual state is held fixed. The
 151 Bregman step gives a closed-form EMA update for that state without additional solver executions.
 152 The practical update follows the finite- K data gradient to the extent that $\hat{f}_{x,t-1}$ tracks $\bar{f}_\phi(x)$.

Table 1: **FB achieves the highest selected accuracy and Pass@10 on Sudoku-Extreme and Maze-Hard.** Selected accuracy uses each method’s native selector, while Pass@10 measures exact solution coverage among ten candidates. Bold and underlined values mark the best and second-best completed results. [†] All HRM values are paper-reported.

Method	Sudoku-Extreme		Maze-Hard		ARC-AGI-2		
	Acc.	Pass@10	Acc.	Pass@10	Pass@1	Pass@2	Pass@100
HRM [†]	55.00	–	74.50	–	–	5.00	–
FPRM	94.20	–	86.90	–	–	–	–
GRAM	98.48	98.49	87.30	87.30	–	–	–
TRM	87.28	–	87.00	–	2.92	4.58	10.83
PTRM	97.36	97.38	<u>87.70</u>	88.40	2.92	3.75	10.83
W-PTRM	<u>98.83</u>	<u>98.84</u>	87.50	<u>89.80</u>	2.92	3.75	11.25
FB (Ours)	99.53	99.55	88.00	91.00	pending	pending	pending

4 Experiments

4.1 Experimental setup

Datasets. We use Sudoku-Extreme, a set of difficult 9×9 Sudoku completion problems, and Maze-Hard, to evaluate optimal path finding in 30×30 grid mazes. For ARC-AGI-2, the model infers an abstract transformation from a small set of input–output grid pairs and applies it to held-out inputs [31].

Baselines. We compare FB with deterministic reasoning models (HRM, TRM, and FPRM), sampling-based reasoning methods (GRAM, PTRM, W-PTRM, and W-FPRM), and posterior optimization methods (IVON and FB). Detailed method descriptions, initialization, and fine-tuning protocols are provided in appendix B.

4.2 Main results: Finite- K posterior learning improves reasoning

Table 1 shows that FB obtains 99.53% selected accuracy and 99.55% Pass@10 on Sudoku-Extreme. These values exceed W-PTRM, the strongest listed comparator for both metrics, by 0.70 and 0.71 percentage points. On Maze-Hard, FB obtains 88.00% selected accuracy and 91.00% Pass@10. The corresponding gains are 0.30 points over PTRM in selected accuracy and 1.20 points over W-PTRM in Pass@10.

The difference between Pass@10 and selected accuracy is 0.02 points on Sudoku-Extreme and 3.00 points on Maze-Hard. Thus, on Maze-Hard, the candidate pool contains an exact solution for more puzzles than the native selector solves. **[RESULT PENDING: We defer the ARC-AGI-2 comparison until the FB evaluation is complete.]**

Candidate coverage under fixed inference budgets. Pass@ K measures how candidate coverage changes as the inference budget grows. We compare FB, ordinary IVON, PTRM, and W-PTRM on the same TRM backbone and Maze-Hard test set at reasoning depth 16. Figure 1a reports Pass@ K at $K \in \{1, 2, 4, 6, 8, 10\}$.

Using all saved FB prefixes to locate the first crossing, FB reaches PTRM’s Pass@10 of 88.4% with two candidates, reducing solver executions by 80%, and reaches W-PTRM’s 89.8% with five candidates, reducing them by 50%. FB also reaches IVON’s Pass@10 of 90.7% with eight candidates, reducing solver executions by 20%; across the displayed budgets, FB exceeds IVON by 0.2–0.4 percentage points.

4.3 Fidelity of one-draw finite- K optimization

We test whether one-draw FB follows the original finite- K objective during training. **[PLACEHOLDER: Compare direct K -draw optimization, the naive one-sample surrogate, and FB**

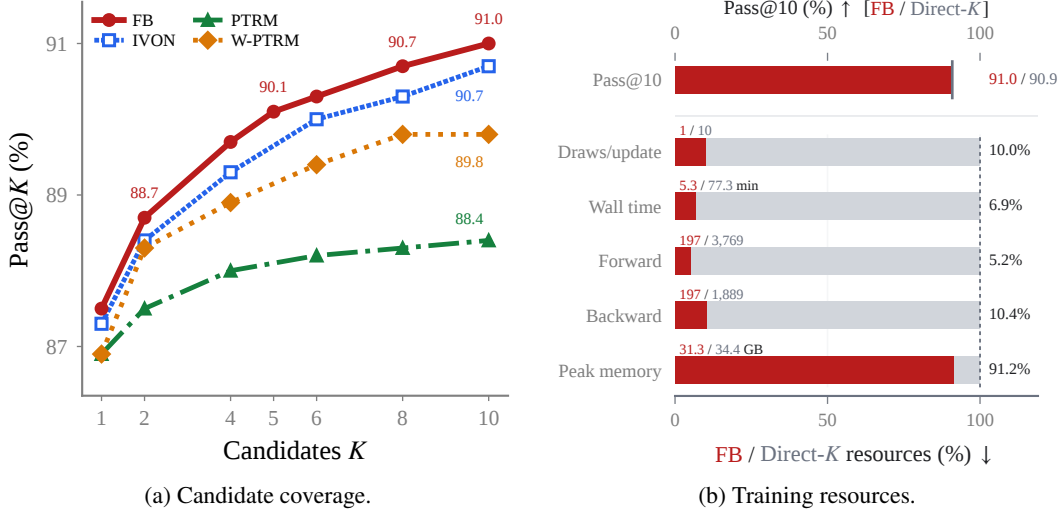


Figure 1: **FB improves inference and training efficiency on Maze-Hard.** (a) FB reaches each baseline’s Pass@10 with fewer candidates using the same TRM backbone, 1,000 test problems, and reasoning depth 16. (b) At 188 updates, one-draw FB maintains comparable Pass@10 to Direct- $K=10$ while reducing training resources. Direct- K uses seed 0; FB reports the mean over seeds 0–2.

185 from matched initialization and training data. Add a three-panel figure reporting (a) the true finite- K
 186 objective estimated with multiple posterior draws, (b) the saddle-objective gap, and (c) dual tracking
 187 error over training.]

188 4.4 Computational advantage of one-draw training

189 Direct finite- K optimization requires K solver executions per update, whereas FB uses one. At
 190 the matched endpoint of 188 updates, Figure 1b compares Direct- $K=10$ seed 0 with the mean of
 191 FB seeds 0–2. FB attains 91.0% Pass@10 versus 90.9% for Direct- K , while using 10.0% as many
 192 posterior draws per update, 6.9% of the wall-clock time, 5.2% of the forward passes, and 10.4% of
 193 the backward passes. Peak memory is reduced more modestly to 91.2%. This endpoint comparison
 194 supports comparable performance with substantially less training compute; it does not establish a
 195 cost-to-target result.

196 5 Limitations

197 6 Conclusion

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A Finite- K and dual-state derivations

A.1 Binary target and direct Monte Carlo estimation

Let $p_\phi(x) = \mathbb{E}_{\theta \sim q_\phi}[F(\theta, x)]$ denote the failure probability of one posterior-sampled solver. For independent draws $\theta_{1:K} \sim q_\phi$,

$$\Pr(\text{all } K \text{ draws fail} \mid x) = \mathbb{E} \left[\prod_{k=1}^K F(\theta_k, x) \right] = p_\phi(x)^K. \quad (8)$$

Hence, the probability of at least one success is $1 - p_\phi(x)^K$, which is the corresponding Pass@ K probability. For binary indicators, the product equals the indicator that all K draws fail. The product therefore represents the deployment event rather than an additional modeling assumption.

Replacing F with the differentiable score f preserves the product across independent draws. Independence gives

$$\mathbb{E} \left[\prod_{k=1}^K f(\theta_k, x) \right] = \prod_{k=1}^K \mathbb{E}[f(\theta_k, x)] = \bar{f}_\phi(x)^K. \quad (9)$$

Thus, the K -draw product is an unbiased estimator of the finite- K data term. Reusing one draw estimates $\mathbb{E}[f(\theta, x)^K]$. For $K > 1$, Jensen’s inequality gives $\mathbb{E}[f^K] \geq \mathbb{E}[f]^K$, with equality only in special cases. In the binary limit, $F^K = F$, so the reused-draw objective reduces to the one-draw failure probability.

A.2 Derivation of the Fenchel representation

For $K \geq 2$, consider $h_r(\lambda) = \lambda r - \psi_K(\lambda)$ on $[0, K]$. Its derivative on $(0, K)$ is

$$h'_r(\lambda) = r - \left(\frac{\lambda}{K} \right)^{1/(K-1)}. \quad (10)$$

The unique stationary point is $\lambda^*(r) = Kr^{K-1}$, which lies in $[0, K]$ for $r \in [0, 1]$. Substitution gives

$$h_r(\lambda^*(r)) = Kr^{K-1}r - (K-1)r^K = r^K. \quad (11)$$

Because ψ_K is strictly convex, h_r is strictly concave and this stationary point is the unique maximizer for $r \in (0, 1)$. The boundary cases give $\lambda^*(0) = 0$ and $\lambda^*(1) = K$. The case $K = 1$ needs no Fenchel step because the objective is linear in r .

A.3 Fixed-dual one-draw gradient

Assume that q_ϕ admits a differentiable reparameterization $\theta = T_\phi(\epsilon)$ whose base distribution does not depend on ϕ . Also assume that an integrable dominating function permits differentiation under the expectation. For a fixed detached dual state λ_x ,

$$\begin{aligned} \nabla_\phi \mathbb{E}_{q_\phi}[\lambda_x f(\theta, x)] &= \nabla_\phi \mathbb{E}_\epsilon[\lambda_x f(T_\phi(\epsilon), x)] \\ &= \mathbb{E}_\epsilon[\nabla_\phi \{\lambda_x f(T_\phi(\epsilon), x)\}]. \end{aligned} \quad (12)$$

One draw of ϵ , or equivalently one $\theta \sim q_\phi$, therefore gives an unbiased stochastic gradient of the fixed-dual data term. Moreover,

$$\nabla_\phi \bar{f}_\phi(x)^K = K \bar{f}_\phi(x)^{K-1} \nabla_\phi \bar{f}_\phi(x) = \lambda^*(\bar{f}_\phi(x)) \nabla_\phi \bar{f}_\phi(x). \quad (13)$$

The fixed-dual gradient coincides with the finite- K data gradient when $\lambda_x = \lambda^*(\bar{f}_\phi(x))$. For a lagged dual state, unbiasedness holds conditionally for the fixed-dual objective.

A.4 Dual coordinate and Bregman update

Lemma 1 (Dual coordinate). *For integer $K \geq 2$ and $\lambda \in [0, K]$, the derivative $\psi'_K(\lambda) = (\lambda/K)^{1/(K-1)}$ is a strictly increasing bijection from $[0, K]$ to $[0, 1]$. Its inverse is $(\psi'_K)^{-1}(u) = Ku^{K-1}$ for $u \in [0, 1]$, and $\psi'_K(\lambda^*(r)) = r$.*

375 *Proof.* Direct differentiation gives the stated derivative, including its continuous extension at zero.
 376 The derivative increases from zero to one on $[0, K]$. Solving $u = (\lambda/K)^{1/(K-1)}$ gives the inverse,
 377 and substituting $\lambda^*(r) = Kr^{K-1}$ gives the final identity. $\square$

378 **Proposition 1** (Bregman-proximal dual update). *Let $K \geq 2$, $\eta > 0$, $\lambda_{x,t} \in [0, K]$, and $f_t \in [0, 1]$.
 379 If $\hat{f}_{x,t-1} = \psi'_K(\lambda_{x,t})$ and $\beta = 1/(1 + \eta)$, then the unique maximizer in eq. (6) is given by eq. (7).*

380 *Proof.* Expanding the Bregman divergence in eq. (6) and dropping terms independent of λ gives

$$Q_t(\lambda) = \left(f_t + \frac{\hat{f}_{x,t-1}}{\eta} \right) \lambda - \left(1 + \frac{1}{\eta} \right) \psi_K(\lambda) + \text{const.} \quad (14)$$

381 Strict convexity of ψ_K makes Q_t strictly concave. The first-order condition gives

$$\psi'_K(\lambda_{x,t+1}) = \frac{\hat{f}_{x,t-1} + \eta f_t}{1 + \eta} = \beta \hat{f}_{x,t-1} + (1 - \beta) f_t. \quad (15)$$

382 The right-hand side lies in $[0, 1]$, so lemma 1 covers the interior and both boundary cases. Applying
 383 $(\psi'_K)^{-1}$ yields $\lambda_{x,t+1} = K \hat{f}_{x,t}^{K-1}$. $\square$

384 A.5 Indexing and the lagged dual state

385 The implementation follows the order

$$\hat{f}_{x,t-1} \longrightarrow \lambda_{x,t} = K \hat{f}_{x,t-1}^{K-1} \longrightarrow \theta_t \sim q_{\phi_t} \longrightarrow f_{x,t} \longrightarrow \hat{f}_{x,t}. \quad (16)$$

386 Let $\mathcal{H}_{t-1}$ denote the training history before drawing θ_t . The state $\lambda_{x,t}$ is measurable with respect
 387 to $\mathcal{H}_{t-1}$ and is fixed before the current posterior sample. The one-draw gradient result therefore
 388 applies conditionally on this history. Detaching a dual state formed from the current sample blocks
 389 its autograd path but does not make it statistically independent of that sample. Using the lagged state
 390 avoids this dependence.

391 A.6 Stationary stochastic tracking

392 **Proposition 2** (Stationary stochastic tracking). *Fix $K \geq 2$ and $\beta \in (0, 1)$. Let $f_t \in [0, 1]$ satisfy
 393 $\mathbb{E}[f_t \mid \mathcal{H}_{t-1}] = r$ and $\mathbb{E}[(f_t - r)^2 \mid \mathcal{H}_{t-1}] \leq \sigma^2$ for fixed $r \in [0, 1]$. For $\hat{f}_t = \beta \hat{f}_{t-1} + (1 - \beta) f_t$,
 394 $\lambda_{t+1} = K \hat{f}_t^{K-1}$, and $\lambda^* = Kr^{K-1}$,*

$$\mathbb{E}[\hat{f}_t] - r = \beta^t (\hat{f}_0 - r), \quad (17)$$

$$\mathbb{E}[(\hat{f}_t - r)^2] \leq \beta^{2t} (\hat{f}_0 - r)^2 + \sigma^2 \frac{1 - \beta}{1 + \beta} (1 - \beta^{2t}), \quad (18)$$

$$\mathbb{E}[(\lambda_{t+1} - \lambda^*)^2] \leq K^2 (K - 1)^2 \mathbb{E}[(\hat{f}_t - r)^2]. \quad (19)$$

395 *The Fenchel inner-max gap $\Delta_t = r^K - [\lambda_{t+1} r - \psi_K(\lambda_{t+1})]$ satisfies*

$$0 \leq \mathbb{E}[\Delta_t] \leq \frac{K(K-1)}{2} \mathbb{E}[(\hat{f}_t - r)^2]. \quad (20)$$

396 *Proof.* Set $e_t = \hat{f}_t - r$ and $\xi_t = f_t - r$. Then $e_t = \beta e_{t-1} + (1 - \beta) \xi_t$ and $\mathbb{E}[\xi_t \mid \mathcal{H}_{t-1}] = 0$.
 397 Taking expectations gives eq. (17). Squaring, conditioning, and unrolling the resulting recursion
 398 gives eq. (18). For $g(u) = Ku^{K-1}$ on $[0, 1]$, the bound $|g'(u)| \leq K(K-1)$ and the mean-value
 399 theorem give eq. (19). Finally, $\psi'_K(\lambda^*) = r$ implies $\Delta_t = D_{\psi_K}(\lambda_{t+1}, \lambda^*)$. Writing $u = \hat{f}_t$ and
 400 integrating the derivative of $r^K - Kr u^{K-1} + (K-1)u^K$ between r and u gives eq. (20). $\square$

A.7 Non-stationary stochastic tracking

Proposition 3 (Non-stationary stochastic tracking). *Fix $K \geq 2$ and $\beta \in (0, 1)$. Let $r_t = \mathbb{E}[f_t \mid \mathcal{H}_{t-1}]$, $\xi_t = f_t - r_t$, and $\mathbb{E}[\xi_t^2 \mid \mathcal{H}_{t-1}] \leq \sigma_t^2$. Define $e_t = \hat{f}_{t-1} - r_t$, $\lambda_t = K\hat{f}_{t-1}^{K-1}$, and $\lambda_t^* = Kr_t^{K-1}$. Then*

$$e_t = \beta^{t-1}e_1 + (1 - \beta) \sum_{j=1}^{t-1} \beta^{t-1-j} \xi_j - \sum_{j=1}^{t-1} \beta^{t-1-j} (r_{j+1} - r_j). \quad (21)$$

Consequently,

$$\begin{aligned} \|e_t\|_2 &\leq \beta^{t-1}\|e_1\|_2 + \sum_{j=1}^{t-1} \beta^{t-1-j} \|r_{j+1} - r_j\|_2 \\ &\quad + (1 - \beta) \left(\sum_{j=1}^{t-1} \beta^{2(t-1-j)} \sigma_j^2 \right)^{1/2}. \end{aligned} \quad (22)$$

If $\|r_{t+1} - r_t\|_2 \leq \delta$ and $\sigma_t \leq \sigma$, then

$$\limsup_{t \rightarrow \infty} \|e_t\|_2 \leq \frac{\delta}{1 - \beta} + \sigma \sqrt{\frac{1 - \beta}{1 + \beta}}. \quad (23)$$

With $\rho_t = \max\{\hat{f}_{t-1}, r_t\}$, the dual error and instantaneous Fenchel gap $\Gamma_t = r_t^K - [\lambda_t r_t - \psi_K(\lambda_t)]$ satisfy

$$|\lambda_t - \lambda_t^*| \leq K(K - 1)\rho_t^{K-2}|e_t|, \quad (24)$$

$$0 \leq \Gamma_t \leq \frac{K(K - 1)}{2} \rho_t^{K-2} e_t^2. \quad (25)$$

Proof. The EMA recursion gives $e_{t+1} = \beta e_t + (1 - \beta)\xi_t - (r_{t+1} - r_t)$. Unrolling it yields eq. (21). The martingale-difference property removes cross terms in the squared norm of the noise sum. The triangle inequality in L_2 then gives eq. (22). Bounding the drift and noise terms by geometric series gives eq. (23). Factoring the difference between $\hat{f}_{t-1}^{K-1}$ and r_t^{K-1} gives eq. (24). The identity $\Gamma_t = D_{\psi_K}(\lambda_t, \lambda_t^*)$ and the same integral argument as in proposition 2 give eq. (25). $\square$

The bounds separate sampling noise from drift in the posterior failure rate. Increasing β reduces the noise term but increases the lag term $\delta/(1 - \beta)$. These results guarantee tracking of the dual state, not convergence of the full neural-network optimization.

B Experimental details

B.1 Datasets and evaluation protocols

[PLACEHOLDER: Report dataset splits, preprocessing, reasoning depth, selectors, the distinction between K_{train} and K_{eval} , evaluation seeds, and confidence-interval construction for Sudoku-Extreme, Maze-Hard, and ARC-AGI-2.]

Maze-Hard candidate-coverage protocol. We evaluate FB, ordinary IVON, PTRM, and W-PTRM on the same 1,000-problem test split with the TRM backbone, reasoning depth 16, and candidate budgets up to ten. FB, PTRM, and W-PTRM use ordered prefixes from saved $K_{\text{eval}} = 10$ banks under seed-0 reset sampling. The reported IVON values use the updated evaluation at $K \in \{1, 2, 4, 6, 8, 10\}$. The earlier IVON bank produced on August 26, 2026 is deprecated and excluded from the reported comparison.

B.2 Baseline details

Deterministic reasoning models. **HRM** couples recurrent modules that operate at different time scales [32]. **TRM** replaces the hierarchy with one small recursive network [33]. **FPRM** uses fixed-point convergence to halt a looped Transformer [34].

Table 2: **FB obtains the highest selected accuracy and Pass@10 with both backbones on Maze-Hard.** Results are percentages on the complete 1,000-puzzle test set. Bold and underlined values mark the best and second-best results within each backbone.

Method	Acc. $\uparrow$	Pass@10 $\uparrow$
<i>TRM backbone</i>		
Deterministic	87.00	87.00
PTRM	<u>87.70</u>	88.40
W-PTRM	87.50	<u>89.80</u>
FB (Ours)	88.00	91.00
<i>FPRM backbone</i>		
Deterministic	<u>86.90</u>	—
W-FPRM	<u>86.90</u>	88.70
IVON	84.40	<u>94.00</u>
FB (Ours)	87.70	94.80

Sampling-based reasoning methods. GRAM models reasoning as stochastic latent trajectories and samples several trajectories in parallel [22]. PTRM keeps the model parameters fixed, injects Gaussian noise into the latent state at each deep-recursion step, and selects among the resulting trajectories [23]. W-PTRM and W-FPRM are non-learned baselines that sample each candidate’s parameters from a fixed Gaussian centered at the released TRM or FPRM checkpoint. They add no latent noise.

Posterior optimization methods. IVON (Improved Variational Online Newton) maintains a sampleable diagonal Gaussian through an online variational update and optimizes the single-draw posterior objective [19]. **FB (Ours)** uses the same IVON posterior parameterization and initialization, but optimizes the finite- K objective with Fenchel–Bregman updates. For learned posterior methods, we initialize the posterior mean from a released task checkpoint and fine-tune its mean and diagonal precision on the task training split. Fine-tuning adapts the solver distribution to the chosen objective, while W-PTRM and W-FPRM isolate sampling without this adaptation.

B.3 Training and compute accounting

[PLACEHOLDER: Report optimization hyperparameters, posterior and dual-state initialization, training-update budgets, hardware, batch sizes, mixed-precision settings, solver-call accounting, wall-clock measurement, and peak-memory measurement.]

Ordinary-IVON baseline. We report the updated IVON Pass@ K evaluation designated on August 27, 2026. The previous local HPO, checkpoint, and candidate bank are deprecated and are not used for the reported curve.

C Additional posterior analyses

C.1 Posterior learning across recursive backbones

Table 2 compares posterior methods with TRM and FPRM backbones on the complete Maze-Hard test set. Every stochastic method uses ten candidates ($K = 10$) and its native selector.

FB improves both metrics under the two tested backbones. With TRM, it raises selected accuracy by 0.30 percentage points over PTRM and Pass@10 by 1.20 points over W-PTRM. With FPRM, FB improves selected accuracy from 84.40% to 87.70% and Pass@10 from 94.00% to 94.80% over IVON. The paired 95% bootstrap intervals are [1.60, 5.10] for selected accuracy and [0.00, 1.60] for Pass@10. The two FPRM posterior rows use the same initialization, training data, update budget, and evaluation noise.

C.2 Posterior construction under a fixed backbone

Table 3 fixes the FPRM backbone and Maze-Hard protocol while varying how each method constructs its parameter distribution. Diagonal Laplace fits a local Gaussian from curvature at a point estimate [20], while SWAG estimates a Gaussian from an SGD trajectory [21].

Table 3: The comparison fixes FPRM and Maze-Hard while varying the posterior construction.

Method	Acc. $\uparrow$	Pass@10 $\uparrow$
FPRM	86.90	–
W-FPRM	86.90	88.70
Diagonal Laplace	pending	pending
SWAG	pending	pending
IVON	84.40	94.00
FB (Ours)	87.70	94.80

Among posterior results, FB improves selected accuracy by 3.30 percentage points and Pass@10 by 0.80 points over IVON. [RESULT PENDING: Compare FB with Diagonal Laplace and SWAG after their matched Maze-Hard results and fit times are available.]

Candidate diversity. [PLACEHOLDER: Report the unique-solution ratio and candidate disagreement for each posterior method. Pair each diversity statistic with Pass@ K so that output variation is not interpreted as useful coverage by itself.]

D Additional objective and resource analyses

D.1 Extended objective and tracking diagnostics

[PLACEHOLDER: Report the naive one-sample surrogate, detailed saddle-objective gap, per-problem and aggregate dual tracking errors, and sensitivity of the tracking gap over training.]

D.2 Extended inference-budget results

[PLACEHOLDER: Report all saved prefixes for $K = 1, \dots, 10$, per-seed results, confidence intervals, and additional sampling baselines that do not fit in the main figure.]

E Ablation studies

EMA coefficient. [PLACEHOLDER: Evaluate $\beta \in \{0, 0.9, 0.99, 0.999\}$ using dual tracking error, the final finite- K objective, optimization stability, and Pass@ K . Relate smaller β to responsive but noisy estimates and larger β to smoother but slower tracking.]

Other optimization choices. [PLACEHOLDER: Report ablations for the KL coefficient τ_{KL} , dual-state initialization, and K_{train} . Include the final finite- K objective, tracking error, Pass@ K , and selected accuracy where applicable.]